

CLEAN DATA FILE

1. แก้ไขข้อมูลผิดรูปแบบ (Inconsistent Data) ปัญหาจากตาราง couriers, products, risk_scores, promotions และ orders โดยกำหนดชนิดข้อมูลเป็น NUMERIC เพราะค่า double precision ถ้าต้องการเอาค่าไปคำนวณอาจจะเกิด Floating Point Error เช่น ทศนิยมบวกกันจาก $0.10 + 0.15 = 0.250000000000002$

BEFORE

	courier_id text	courier_name text	courier_type text	avg_delivery_days double precision	damage_rate double precision	coverage_region text
1	COUR01	FastShip	Express	1.5	0.01	Nationwide
2	COUR02	SafeLogistics	Standard	3	0.02	Nationwide
3	COUR03	EcoDelivery	Eco	5	0.05	Bangkok Only

channel_type text	payment_method text	quantity bigint	unit_price double precision	tier_discount_pct double precision	campaign_discount_pct double precision	total_discount_pct double precision	discount_applied_amount double precision	total_amount double precision
Shopee	Bank_Transfer	1	2500	0.05	0.1	0.15000000000000002	375.00000000000006	
TikTok	COD	2	2500	0.05	0.1	0.15000000000000002	750.00000000000001	
Shopee	Bank_Transfer	2	450	0.05	0	0.05	45	
TikTok	COD	1	1200	0.05	0	0.05	60	
Mobile_App	COD	2	1500	0.05	0	0.05	150	
TikTok	Bank_Transfer	2	1500	0.1	0.1	0.2	600	
TV_Show	Bank_Transfer	2	990	0.05	0	0.05	99	
Mobile_App	COD	1	1500	0.1	0	0.1	150	
TikTok	COD	1	650	0.2	0	0.2	130	
TikTok	Credit_Card	1	1100	0.05	0	0.05	55	
Mobile_App	COD	2	1200	0.05	0	0.05	120	
Shopee	Bank_Transfer	1	1800	0.1	0.1	0.2	360	
Shopee	Credit_Card	1	990	0.1	0.1	0.2	198	

total_discount_pct double precision	discount_applied_amount double precision	total_amount double precision	delivery_days bigint	is_repurchased_item bigint	order_hour bigint	hist_return_rate double precision
0.05	180	3420	4	0	15	0
0.25	175	525	2	0	19	0
0.2	880	3520	6	0	21	1
0.05	99	1881	5	0	21	0
0.05	250	4750	6	0	21	0
0.05	125	2375	6	1	1	1
0.05	44.5	845.5	5	0	14	0
0.1	58	522	2	0	17	0
0.15	180	1020	4	0	19	0.3333333333333333
0.25	1750	5250	1	0	22	0
0.05	120	2280	4	0	23	1
0.2	130	520	3	0	4	0

unit_price double precision	is_fragile boolean	product_rating double precision
290	false	4.5
890	false	4.2
1990	false	4.8
2500	true	3.5
1200	true	4
3500	true	4.7
990	true	3.8
350	false	4.4
2200	true	4.1
1100	true	3.9
4500	true	4.6
1500	true	4.3
450	true	4.9

	promo_id text	promo_name text	promo_type text	discount_rate double precision	start_date timestamp without time zone	end_date timestamp without time zone
1	PROMO_001	New Year Mega Sale	Campaign	0.1	2025-01-01 00:00:00	2025-01-31 00:00:00
2	PROMO_002	Summer Fashion We...	Campaign	0.05	2025-04-01 00:00:00	2025-04-30 00:00:00
3	PROMO_003	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00
4	PROMO_NONE	No Promotion	None	0	2020-01-01 00:00:00	2030-01-01 00:00:00

ดังนั้นในส่วนนี้เราจึงต้องปรับแก้ข้อมูลจากทศนิยมหลายๆ ตำแหน่งให้เหลือแค่ทศนิยม 2 ตำแหน่งเพราะ
 ในฐานข้อมูลของเราใช้ทศนิยมแค่ 2 ตำแหน่งเท่านั้น

วิธีการแก้ไขหรือการ Cleansing ข้อมูล

1.

```
ALTER TABLE couriers
```

```
ALTER COLUMN avg_delivery_days TYPE NUMERIC(10, 2);
```

```
ALTER TABLE couriers
```

```
ALTER COLUMN damage_rate TYPE NUMERIC(10, 2);
```

2.

ALTER TABLE orders

ALTER COLUMN unit_price TYPE NUMERIC(10, 2);

ALTER TABLE orders

ALTER COLUMN tier_discount_pct TYPE NUMERIC(10, 2);

ALTER TABLE orders

ALTER COLUMN campaign_discount_pct TYPE NUMERIC(10, 2);

3.

ALTER TABLE orders

ALTER COLUMN total_discount_pct TYPE NUMERIC(10, 2);

ALTER TABLE orders

ALTER COLUMN discount_applied_amount TYPE NUMERIC(10, 2);

ALTER TABLE orders

ALTER COLUMN total_amount TYPE NUMERIC(10, 2);

4.

ALTER TABLE orders

ALTER COLUMN hist_return_rate TYPE NUMERIC(10, 2);

5.

ALTER TABLE products

ALTER COLUMN unit_price TYPE NUMERIC(10, 2);

6.

ALTER TABLE products

ALTER COLUMN product_rating TYPE NUMERIC(10, 2);

AFTER (Result)

1.

	courier_id [PK] text	courier_name text	courier_type text	avg_delivery_days numeric (10,2)	damage_rate numeric (10,2)	coverage_region text
1	COUR01	FastShip	Express	1.50	0.01	Nationwide
2	COUR02	SafeLogistics	Standard	3.00	0.02	Nationwide
3	COUR03	EcoDelivery	Eco	5.00	0.05	Bangkok Only

2.

unit_price numeric (10,2)	tier_discount_pct numeric (10,2)	campaign_discount_pct numeric (10,2)
2500.00	0.05	0.10
2500.00	0.05	0.10
450.00	0.05	0.00
1200.00	0.05	0.00
1500.00	0.05	0.00
1500.00	0.10	0.10
990.00	0.05	0.00
1500.00	0.10	0.00
650.00	0.20	0.00
1100.00	0.05	0.00
1200.00	0.05	0.00
1800.00	0.10	0.10

3.

total_discount_pct numeric (10,2)	discount_applied_amount numeric (10,2)	total_amount numeric (10,2)
0.15	375.00	2125.00
0.15	750.00	4250.00
0.05	45.00	855.00
0.05	60.00	1140.00
0.05	150.00	2850.00
0.20	600.00	2400.00
0.05	99.00	1881.00
0.10	150.00	1350.00
0.20	130.00	520.00
0.05	55.00	1045.00
0.05	120.00	2280.00
0.20	360.00	1440.00

4.

hist_return_rate
numeric (10,2)
0.00
0.00
1.00
0.00
0.00
1.00
0.00
0.00
0.33
0.00
1.00
0.00
0.00

5.

unit_price	is_fragile	product_rating
numeric (10,2)	boolean	numeric (10,2)
290.00	false	4.50
890.00	false	4.20
1990.00	false	4.80
2500.00	true	3.50
1200.00	true	4.00
3500.00	true	4.70
990.00	true	3.80
350.00	false	4.40
2200.00	true	4.10
1100.00	true	3.90
4500.00	true	4.60
1500.00	true	4.30
450.00	true	4.90

6.

	promo_id	promo_name	promo_type	discount_rate	start_date	end_date
	text	text	text	numeric (10,2)	timestamp without time zone	timestamp without time zone
1	PROMO_001	New Year Mega Sale	Campaign	0.10	2025-01-01 00:00:00	2025-01-31 00:00:00
2	PROMO_002	Summer Fashion We...	Campaign	0.05	2025-04-01 00:00:00	2025-04-30 00:00:00
3	PROMO_003	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00
4	PROMO_NONE	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00

2. ค่าว่าง (Missing Values) จากตาราง order_history_raw_data หลังจากการ join table ทั้งหมด

BEFORE

return_id text	customer_id_ret text	return_date timestamp without time zone	return_reason text	return_scenario text	item_condition text	return_status text	refund_amount double precision
RET00001	C0111	2025-01-08 05:00:00	Changed Mind	Standard Return	Defective	Completed	2125
RET00002	C0300	2025-01-09 08:00:00	Better Price Elsewh...	Standard Return	Defective	Completed	4250
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]
[null]	[null]	[null]	[null]	[null]	[null]	[null]	[null]

วิธีการแก้ไขหรือการ Cleansing ข้อมูล

พอเรา Join ตาราง return มาแล้วค่ามันหายไป เราเลยต้องจัดการแทนที่ค่าที่หายไปก็จะมี Field เหล่านี้

1. เติม 'None' สำหรับรหัสการคืนที่ว่าง (return_id)
2. เติม 'Not Returned' สำหรับเหตุผลและสถานะต่างๆ (return_reason, return_scenario, item_condition, return_status)
3. เติม 1900-01-01 ใช้วันที่สมมติเพื่อให้ไม่มีค่าว่าง (return_date)
4. เติม 0.00 จัดการเรื่องตัวเลข (refund_amount)

```
UPDATE order_history_raw_data
```

```
SET
```

```
    return_id = COALESCE(return_id, 'None'),
```

```
    return_reason = COALESCE(return_reason, 'Not Returned'),
```

```
    return_scenario = COALESCE(return_scenario, 'Not Returned'),
```

```
    item_condition = COALESCE(item_condition, 'Not Returned'),
```

```
    return_status = COALESCE(return_status, 'Not Returned'),
```

refund_amount = COALESCE(refund_amount, 0.00)

return_date = COALESCE(return_date, '1900-01-01')

WHERE is_returned = 0;

AFTER (Result)

return_id text	customer_id_ret text	return_date timestamp without time zone	return_reason text	return_scenario text	item_condition text	return_status text	refund_amount numeric (10,2)
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00
None	[null]	1900-01-01 00:00:00	Not Returned	Not Returned	Not Returned	Not Returned	0.00

3. กำจัดข้อมูลซ้ำซ้อนและไม่เกี่ยวข้อง (Irrelevant/Redundant) จากตาราง order_history_raw_data
หลังจากการ join table ทั้งหมด

BEFORE

customer_id_ret text	unit_price_y double precision	unit_price_x double precision
[null]	2200	2200
[null]	1200	1200
[null]	450	450
[null]	2800	2800
[null]	1500	1500
[null]	4500	4500
[null]	2500	2500
[null]	1200	1200
[null]	2800	2800
[null]	750	750
[null]	990	990
[null]	450	450
[null]	650	650
[null]	3500	3500

วิธีการแก้ไขหรือการ Cleansing ข้อมูล

พอเรา Join ตารางมาแล้วมันจะมี Column ที่ซ้ำกัน และ Column ที่ไม่จำเป็นดังนั้นเราจึงต้องจัดการประมาณนี้ เนื่องจากว่ามันมาจากหลายๆ ตารางทำให้อาจจะมี Column ที่ซ้ำกันเกิดขึ้น โดย

- DROP Column ที่ไม่ใช่ 1 ตารางก็คือ unit_price_y ให้เหลือแค่ unit_price_x แล้วเปลี่ยนชื่อ Column เป็น unit_price (ไว้เก็บราคาของสินค้า)
- DROP Column customer_id_ret ที่ซ้ำกับ customer_id

คำสั่ง SQL Query

```
ALTER TABLE order_history_rawdata DROP COLUMN IF EXISTS unit_price_y;
```

```
ALTER TABLE order_history_rawdata DROP COLUMN IF EXISTS customer_id_ret;
```

```
ALTER TABLE order_history_rawdata RENAME COLUMN unit_price_x TO unit_price;
```

AFTER (Result)

	return_id text	return_date timestamp without time zone	return_reason text	return_scenario text	item_condition text	return_status text
0...	RET00001	2025-01-08 05:00:00	Changed Mind	Standard Return	Defective	Completed
0...	RET00002	2025-01-09 08:00:00	Better Price Elsewh...	Standard Return	Defective	Completed
0...	RET00003	2025-01-08 20:00:00	Defective	Standard Return	Unopened	Completed
0...	RET00004	2025-01-12 04:00:00	Defective	Standard Return	Unopened	Completed
0...	RET00005	2025-01-10 08:00:00	Better Price Elsewh...	Standard Return	Damaged Packagi...	Completed
0...	RET00006	2025-01-13 15:00:00	Changed Mind	Standard Return	Defective	Completed
0...	RET00007	2025-01-16 21:00:00	Defective	Standard Return	Used	Completed
0...	RET00008	2025-01-17 01:00:00	Wrong Item	Standard Return	Unopened	Completed
0...	RET00009	2025-01-16 17:00:00	Wrong Item	Standard Return	Used	Completed
0...	RET00010	2025-01-16 22:00:00	Changed Mind	Standard Return	Damaged Packagi...	Completed
0...	RET00011	2025-01-22 21:00:00	Wrong Item	Standard Return	Damaged Packagi...	Completed
0...	RET00012	2025-01-15 19:00:00	Wrong Item	Standard Return	Damaged Packagi...	Completed
0...	RET00013	2025-01-19 04:00:00	Changed Mind	Standard Return	Defective	Completed
0...	RET00014	2025-01-16 12:00:00	Better Price Elsewh...	Standard Return	Used	Completed
0...	RET00015	2025-01-23 00:00:00	Wrong Item	Standard Return	Used	Completed

	unit_price numeric (10,2)	tier_discount_pct numeric (10,2)	campaign_discount_pct numeric (10,2)	total_discount_pct numeric (10,2)	discount_applied_amount numeric (10,2)	total_amount numeric (10,2)	delivery_days bigint	is_repurcha bigint
1	2500.00	0.05	0.10	0.15	375.00	2125.00	4	
2	2500.00	0.05	0.10	0.15	750.00	4250.00	3	
2	990.00	0.05	0.00	0.05	99.00	1881.00	2	
2	3500.00	0.05	0.10	0.15	1050.00	5950.00	6	
1	290.00	0.05	0.10	0.15	43.50	246.50	3	
2	1800.00	0.05	0.00	0.05	180.00	3420.00	4	
2	2500.00	0.05	0.00	0.05	250.00	4750.00	6	
1	2500.00	0.05	0.00	0.05	125.00	2375.00	6	
2	4500.00	0.05	0.00	0.05	450.00	8550.00	5	
2	1500.00	0.05	0.00	0.05	150.00	2850.00	6	
1	2500.00	0.10	0.00	0.10	250.00	2250.00	6	
2	890.00	0.05	0.00	0.05	89.00	1691.00	1	
2	890.00	0.05	0.00	0.05	89.00	1691.00	5	
1	890.00	0.10	0.00	0.10	89.00	801.00	5	
2	1200.00	0.10	0.00	0.10	240.00	2160.00	5	

4. แก้ไขทศนิยมเพี้ยน (Inconsistent Data - Floating Point) จากตาราง order_history_rawdata หลังจากการ join table ทั้งหมดเข้ามา

BEFORE

unit_price double precision	tier_discount_pct double precision	campaign_discount_pct double precision	total_discount_pct double precision	discount_applied_amount double precision	total_amount double precision
2500	0.05	0.1	0.15000000000000002	375.00000000000006	2125
2500	0.05	0.1	0.15000000000000002	750.00000000000001	4250
990	0.05	0	0.05	99	1881
3500	0.05	0.1	0.15000000000000002	1050.00000000000002	5950
290	0.05	0.1	0.15000000000000002	43.500000000000001	246.5
1800	0.05	0	0.05	180	3420
2500	0.05	0	0.05	250	4750
2500	0.05	0	0.05	125	2375
4500	0.05	0	0.05	450	8550
1500	0.05	0	0.05	150	2850
2500	0.1	0	0.1	250	2250
890	0.05	0	0.05	89	1691

วิธีการแก้ไขหรือการ Cleansing ข้อมูล

จากข้อมูลเราจะเห็นว่ามีความค่าของตัวเลขทศนิยมหลายๆ ตำแหน่งมากเกินไปดังนั้นเวลาที่เราจะเอาค่าออกไปคำนวณต่อทำให้อาจจะเกิดผิดพลาดหรือค่าเพี้ยนไปจากเดิมเราจึงต้องจัดการปัญหาเหล่านี้โดยใช้ ALTER และแปลงชนิดข้อมูล Column เช่น TABLE order_history_rawdata

```
ALTER TABLE order_history_rawdata
```

```
ALTER COLUMN unit_price TYPE NUMERIC(10, 2);
```

AFTER (Result)

unit_price numeric (10,2)	tier_discount_pct numeric (10,2)	campaign_discount_pct numeric (10,2)	total_discount_pct numeric (10,2)	discount_applied_amount numeric (10,2)	total_amount numeric (10,2)	delivery_days bigint
2500.00	0.05	0.10	0.15	375.00	2125.00	4
2500.00	0.05	0.10	0.15	750.00	4250.00	3
990.00	0.05	0.00	0.05	99.00	1881.00	2
3500.00	0.05	0.10	0.15	1050.00	5950.00	6
290.00	0.05	0.10	0.15	43.50	246.50	3
1800.00	0.05	0.00	0.05	180.00	3420.00	4
2500.00	0.05	0.00	0.05	250.00	4750.00	6
2500.00	0.05	0.00	0.05	125.00	2375.00	6
4500.00	0.05	0.00	0.05	450.00	8550.00	5
1500.00	0.05	0.00	0.05	150.00	2850.00	6
2500.00	0.10	0.00	0.10	250.00	2250.00	6
890.00	0.05	0.00	0.05	89.00	1691.00	1
890.00	0.05	0.00	0.05	89.00	1691.00	5
890.00	0.10	0.00	0.10	89.00	801.00	5
1200.00	0.10	0.00	0.10	240.00	2160.00	5

ค่าที่รันด้วยข้อมูลดิบออกมาดูตัวอย่างก่อน (order_history_raw_data)

	order_id text	customer_id text	product_id text	courier_id text	promo_id text	order_date text	expected_delivery_date text	delivery_date text	delivery_time_expected_days bigint
1	ORD000...	C0111	PRD004	COUR01	PROMO_001	2025-01-01 05:00:...	2025-01-02 05:00:00	2025-01-05 05:00:...	1
2	ORD000...	C0300	PRD004	COUR02	PROMO_001	2025-01-01 08:00:...	2025-01-03 08:00:00	2025-01-04 08:00:...	2
3	ORD000...	C0224	PRD013	COUR01	PROMO_NO...	2025-01-01 10:00:...	2025-01-03 10:00:00	2025-01-07 10:00:...	2
4	ORD000...	C0463	PRD005	COUR01	PROMO_NO...	2025-01-01 13:00:...	2025-01-04 13:00:00	2025-01-03 13:00:...	3
5	ORD000...	C0384	PRD012	COUR01	PROMO_NO...	2025-01-01 14:00:...	2025-01-03 14:00:00	2025-01-03 14:00:...	2
6	ORD000...	C0464	PRD012	COUR02	PROMO_001	2025-01-01 16:00:...	2025-01-02 16:00:00	2025-01-07 16:00:...	1
7	ORD000...	C0350	PRD007	COUR03	PROMO_NO...	2025-01-01 19:00:...	2025-01-02 19:00:00	2025-01-02 19:00:...	1
8	ORD000...	C0129	PRD012	COUR02	PROMO_NO...	2025-01-01 21:00:...	2025-01-03 21:00:00	2025-01-02 21:00:...	2
9	ORD000...	C0470	PRD014	COUR01	PROMO_NO...	2025-01-01 22:00:...	2025-01-03 22:00:00	2025-01-06 22:00:...	2
10	ORD000...	C0399	PRD010	COUR02	PROMO_NO...	2025-01-02 07:00:...	2025-01-05 07:00:00	2025-01-04 07:00:...	3
11	ORD000...	C0287	PRD019	COUR01	PROMO_NO...	2025-01-02 11:00:...	2025-01-05 11:00:00	2025-01-03 11:00:...	3
12	ORD000...	C0480	PRD017	COUR01	PROMO_001	2025-01-02 19:00:...	2025-01-05 19:00:00	2025-01-06 19:00:...	3
13	ORD000...	C0172	PRD007	COUR02	PROMO_001	2025-01-02 22:00:...	2025-01-05 22:00:00	2025-01-03 22:00:...	3
14	ORD000...	C0260	PRD001	COUR03	PROMO_001	2025-01-03 08:00:...	2025-01-05 08:00:00	2025-01-05 08:00:...	2
15	ORD000...	C0236	PRD009	COUR02	PROMO_001	2025-01-03 10:00:...	2025-01-04 10:00:00	2025-01-05 10:00:...	1

channel_type text	payment_method text	quantity bigint	unit_price_x double precision	tier_discount_pct double precision	campaign_discount_pct double precision	total_discount_pct double precision	discount_applied_amount double precision
Shopee	Bank_Transfer	1	2500	0.05	0.1	0.15	375.00000000000006
TikTok	COD	2	2500	0.05	0.1	0.15	750.00000000000001
Shopee	Bank_Transfer	2	450	0.05	0	0.05	45
TikTok	COD	1	1200	0.05	0	0.05	60
Mobile_App	COD	2	1500	0.05	0	0.05	150
TikTok	Bank_Transfer	2	1500	0.1	0.1	0.2	600
TV_Show	Bank_Transfer	2	990	0.05	0	0.05	99
Mobile_App	COD	1	1500	0.1	0	0.1	150
TikTok	COD	1	650	0.2	0	0.2	130
TikTok	Credit_Card	1	1100	0.05	0	0.05	55
Mobile_App	COD	2	1200	0.05	0	0.05	120
Shopee	Bank_Transfer	1	1800	0.1	0.1	0.2	360
Shopee	Credit_Card	1	990	0.1	0.1	0.2	198
Mobile_App	Bank_Transfer	2	290	0.05	0.1	0.15	87.000000000000001
TV_Show	Bank_Transfer	2	2200	0.05	0.1	0.15	660.00000000000001

total_amount double precision	delivery_days bigint	is_repurchased_item bigint	order_hour bigint	days_since_last_order bigint	hist_order_count bigint	hist_return_rate double precision	is_returned bigint	customer_name text
2125	4	0	5	-1	0	0	1	Anong Suksamran
4250	3	0	8	-1	0	0	1	Kasem Srisai
855	6	0	10	-1	0	0	0	Somsak Rak-Thai
1140	2	0	13	-1	0	0	0	Nipa Sawasdee
2850	2	0	14	-1	0	0	0	Somsak Thong-In
2400	6	0	16	-1	0	0	0	Siri Rak-Thai
1881	1	0	19	-1	0	0	0	Wichai Jaidee
1350	1	0	21	-1	0	0	0	Siri Sawasdee
520	5	0	22	-1	0	0	0	Anong Suksamran
1045	2	0	7	-1	0	0	0	Siri Rattanakul
2280	1	0	11	-1	0	0	0	Sunee Rak-Thai
1440	4	0	19	-1	0	0	0	Somsak Mungkorn
792	1	0	22	-1	0	0	0	Wichai Jaidee
493	2	0	8	-1	0	0	0	Kasem Srisai
3740	2	0	10	-1	0	0	0	Malee Srisai

customer_phone text	membership_tier text	preferred_channel text	province text	registration_date text	customer_age_days bigint	product_name text	category text	brand text
083-875-3708	Bronze	App	Bangkok	2022-02-21	1542	Silk Dress	Fashion	SilkTouch
082-943-4432	Bronze	Web	Khon Kaen	2023-05-01	1108	Silk Dress	Fashion	SilkTouch
081-756-4676	Bronze	Web	Nonthaburi	2020-12-29	1961	Matte Lipstick	Cosmetics	GlowUp
084-919-4292	Bronze	LINE	Phuket	2019-07-14	2495	Wireless Earbuds	Electronics	SoundMax
081-245-6010	Bronze	TV	Nonthaburi	2022-02-22	1541	Rice Cooker	Home_Applian...	CookEasy
084-519-5517	Silver	TV	Chiang Mai	2020-02-13	2281	Rice Cooker	Home_Applian...	CookEasy
086-366-8010	Bronze	LINE	Nonthaburi	2024-05-24	719	Bluetooth Speak...	Electronics	BoomBox
082-282-8126	Silver	Web	Bangkok	2024-05-18	725	Rice Cooker	Home_Applian...	CookEasy
086-830-7548	Platinum	LINE	Chonburi	2022-10-17	1304	Sunscreen SPF50	Cosmetics	SunSafe
083-172-6472	Bronze	Web	Bangkok	2024-07-16	666	Blender	Home_Applian...	BlendIt
086-101-1374	Bronze	TV	Bangkok	2023-10-23	933	Collagen Drink	Supplement	GlowDrink
084-867-8961	Silver	App	Chonburi	2024-11-11	548	Whey Protein	Supplement	MuscleF...
085-275-9228	Silver	TV	Phuket	2022-02-09	1554	Bluetooth Speak...	Electronics	BoomBox
086-413-1021	Bronze	TV	Khon Kaen	2020-08-02	2110	Cotton T-Shirt	Fashion	UniStyle
088-816-5167	Bronze	App	Khon Kaen	2022-01-22	1572	Microwave	Home_Applian...	HomePro

supplier_name text	contact text	promo_name text	promo_type text	discount_rate double precision	start_date text	end_date text	score_id text	risk_score double precision
Thai Fashion Co.	02-202-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00001	0.3399999999999999
Thai Fashion Co.	02-202-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00002	0.39
Beauty Lab Inc.	02-370-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00003	0.19
BangkokTech L...	02-535-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00004	0.04
HomeStyle Gro...	02-960-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00005	0.04
HomeStyle Gro...	02-960-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00006	0.19
GadgetWorld	02-120-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00007	0.19
HomeStyle Gro...	02-960-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00008	0.04
Beauty Lab Inc.	02-370-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00009	0.19
HomeStyle Gro...	02-960-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00010	0.19
NatureCare	02-221-00...	No Promotion	[null]	0	2020-01-01	2030-01-01	SCR00011	0.04
Wellness Plus	02-206-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00012	0.19
GadgetWorld	02-120-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00013	0.19
Thai Fashion Co.	02-202-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00014	0.04
HomeStyle Gro...	02-960-00...	New Year Mega Sale	Campaign	0.1	2025-01-01	2025-01-31	SCR00015	0.24

supplier_id text	unit_price_y double precision	is_fragile boolean	product_rating double precision	courier_name text	courier_type text	avg_delivery_days double precision	damage_rate double precision	coverage_region text
SUP001	2500	true	3.5	FastShip	Express	1.5	0.01	Nationwide
SUP001	2500	true	3.5	SafeLogistics	Standard	3	0.02	Nationwide
SUP004	450	true	4.9	FastShip	Express	1.5	0.01	Nationwide
SUP002	1200	true	4	FastShip	Express	1.5	0.01	Nationwide
SUP003	1500	true	4.3	FastShip	Express	1.5	0.01	Nationwide
SUP003	1500	true	4.3	SafeLogistics	Standard	3	0.02	Nationwide
SUP008	990	true	3.8	EcoDelivery	Eco	5	0.05	Bangkok Only
SUP003	1500	true	4.3	SafeLogistics	Standard	3	0.02	Nationwide
SUP004	650	false	4.5	FastShip	Express	1.5	0.01	Nationwide
SUP003	1100	true	3.9	SafeLogistics	Standard	3	0.02	Nationwide
SUP010	1200	false	4.3	FastShip	Express	1.5	0.01	Nationwide
SUP005	1800	false	4	FastShip	Express	1.5	0.01	Nationwide
SUP008	990	true	3.8	SafeLogistics	Standard	3	0.02	Nationwide
SUP001	290	false	4.5	EcoDelivery	Eco	5	0.05	Bangkok Only
SUP003	2200	true	4.1	SafeLogistics	Standard	3	0.02	Nationwide

ค่าที่รันด้วยข้อมูลคลีนดูตัวอย่างหลัง (order_history_clean_data)

	order_id text	customer_id text	product_id text	courier_id text	promo_id text	order_date timestamp without time zone	expected_delivery_date timestamp without time zone	delivery_date timestamp without time zone	
601	ORD022...	C0069	PRD009	COUR02	PROMO_NONE	2025-08-15 04:00:00	2025-08-18 04:00:00	2025-08-21 04:00:00	
602	ORD022...	C0012	PRD007	COUR03	PROMO_003	2025-08-15 11:00:00	2025-08-17 11:00:00	2025-08-20 11:00:00	
603	ORD022...	C0330	PRD020	COUR01	PROMO_NONE	2025-08-15 23:00:00	2025-08-17 23:00:00	2025-08-19 23:00:00	
604	ORD022...	C0344	PRD007	COUR01	PROMO_003	2025-08-16 08:00:00	2025-08-19 08:00:00	2025-08-22 08:00:00	
605	ORD022...	C0214	PRD002	COUR02	PROMO_003	2025-08-16 13:00:00	2025-08-19 13:00:00	2025-08-21 13:00:00	
606	ORD022...	C0231	PRD003	COUR02	PROMO_NONE	2025-08-16 14:00:00	2025-08-17 14:00:00	2025-08-21 14:00:00	
607	ORD022...	C0184	PRD006	COUR03	PROMO_NONE	2025-08-17 20:00:00	2025-08-18 20:00:00	2025-08-18 20:00:00	
608	ORD023...	C0090	PRD008	COUR02	PROMO_NONE	2025-08-18 02:00:00	2025-08-20 02:00:00	2025-08-21 02:00:00	
609	ORD023...	C0244	PRD008	COUR03	PROMO_003	2025-08-18 07:00:00	2025-08-20 07:00:00	2025-08-24 07:00:00	
610	ORD023...	C0170	PRD018	COUR01	PROMO_NONE	2025-08-18 08:00:00	2025-08-20 08:00:00	2025-08-19 08:00:00	
611	ORD023...	C0215	PRD009	COUR01	PROMO_NONE	2025-08-18 12:00:00	2025-08-21 12:00:00	2025-08-23 12:00:00	
612	ORD023...	C0376	PRD002	COUR01	PROMO_NONE	2025-08-19 14:00:00	2025-08-21 14:00:00	2025-08-22 14:00:00	
613	ORD023...	C0044	PRD003	COUR03	PROMO_003	2025-08-19 18:00:00	2025-08-20 18:00:00	2025-08-24 18:00:00	
614	ORD023...	C0095	PRD010	COUR03	PROMO_003	2025-08-19 18:00:00	2025-08-20 18:00:00	2025-08-20 18:00:00	
615	ORD023...	C0250	PRD009	COUR01	PROMO_003	2025-08-19 22:00:00	2025-08-22 22:00:00	2025-08-25 22:00:00	
delivery_time_expected_days bigint		channel_type text	payment_method text	quantity bigint	unit_price numeric (10,2)	tier_discount_pct numeric (10,2)	campaign_discount_pct numeric (10,2)	total_discount_pct numeric (10,2)	
		3	TikTok	COD	1	2200.00	0.05	0.00	0.05
		2	Shopee	COD	1	990.00	0.10	0.15	0.25
		2	TV_Show	Credit_Card	1	750.00	0.05	0.00	0.05
		3	Shopee	COD	1	990.00	0.05	0.15	0.20
		3	Shopee	Credit_Card	2	890.00	0.05	0.15	0.20
		1	TikTok	Credit_Card	2	1990.00	0.15	0.00	0.15
		1	Mobile_App	Credit_Card	2	3500.00	0.15	0.00	0.15
		2	TikTok	Bank_Transfer	1	350.00	0.10	0.00	0.10
		2	TikTok	Credit_Card	2	350.00	0.10	0.15	0.25
		2	Mobile_App	COD	2	890.00	0.10	0.00	0.10
		3	Shopee	Credit_Card	1	2200.00	0.05	0.00	0.05
		2	TikTok	Credit_Card	2	890.00	0.15	0.00	0.15
		1	Mobile_App	COD	1	1990.00	0.10	0.15	0.25
		1	Shopee	Credit_Card	2	1100.00	0.10	0.15	0.25
		3	Shopee	Bank_Transfer	1	2200.00	0.05	0.15	0.20
discount_applied_amount numeric (10,2)		total_amount numeric (10,2)	delivery_days bigint	is_repurchased_item bigint	order_hour bigint	days_since_last_order bigint	hist_order_count bigint	hist_return_rate numeric (10,2)	
		110.00	2090.00	6	1	4	0	7	0.43
		247.50	742.50	5	0	11	52	9	0.67
		37.50	712.50	4	1	23	25	8	0.38
		198.00	792.00	6	0	8	6	3	0.00
		356.00	1424.00	5	0	13	12	11	0.45
		597.00	3383.00	5	0	14	11	9	0.67
		1050.00	5950.00	1	0	20	17	4	0.00
		35.00	315.00	3	0	2	34	5	0.40
		175.00	525.00	6	0	7	58	1	0.00
		178.00	1602.00	1	0	8	5	5	0.00
		110.00	2090.00	5	0	12	63	7	0.29
		267.00	1513.00	3	0	14	53	4	1.00
		497.50	1492.50	5	0	18	102	5	0.20
		550.00	1650.00	1	0	18	2	8	0.63
		440.00	1760.00	6	0	22	176	2	0.00

is_returned bigint	customer_name text	customer_phone text	membership_tier text	preferred_channel text	province text	registration_date timestamp without time zone	customer_age_days bigint		
1	Sunee Suksamran	081-625-3599	Bronze	TV	Nonthaburi	2020-01-22 00:00:00	2303		
1	Kasem Mungkorn	084-412-5212	Silver	Web	Phuket	2021-02-26 00:00:00	1902		
1	Kasem Wongsuw...	088-402-1986	Bronze	TV	Phuket	2020-12-21 00:00:00	1969		
1	Preecha Srisai	086-921-1454	Bronze	LINE	Chiang Mai	2022-06-15 00:00:00	1428		
1	Thana Sawasdee	086-558-6658	Bronze	LINE	Chonburi	2021-08-28 00:00:00	1719		
1	Thana Jaidee	081-833-2875	Gold	Web	Chonburi	2019-09-11 00:00:00	2436		
1	Malee Sawasdee	083-521-5795	Gold	TV	Khon Kaen	2024-03-24 00:00:00	780		
1	Anong Jaidee	083-611-7819	Silver	Web	Songkhla	2024-01-14 00:00:00	850		
1	Nipa Suksamran	087-691-3720	Silver	TV	Chiang Mai	2022-05-10 00:00:00	1464		
1	Somsak Thong-In	082-223-1974	Silver	LINE	Bangkok	2020-08-26 00:00:00	2086		
1	Thana Rak-Thai	083-948-1988	Bronze	TV	Songkhla	2019-08-19 00:00:00	2459		
1	Somsak Wongsu...	086-319-4097	Gold	TV	Chonburi	2024-01-25 00:00:00	839		
1	Preecha Rattana...	081-858-6954	Silver	Web	Bangkok	2022-05-04 00:00:00	1470		
1	Siri Thong-In	087-247-2762	Silver	Web	Phuket	2024-07-02 00:00:00	680		
1	Sunee Pattana	082-986-9619	Bronze	TV	Songkhla	2024-03-24 00:00:00	780		
product_name text	category text	brand text	supplier_id text	is_fragile boolean	product_rating numeric (10,2)	courier_name text	courier_type text	avg_delivery_days numeric (10,2)	damage_rate numeric (10,2)
Microwave	Home_Applian...	HomePro	SUP003	true	4.10	SafeLogistics	Standard	3.00	0.02
Bluetooth Speak...	Electronics	BoomBox	SUP008	true	3.80	EcoDelivery	Eco	5.00	0.05
Fish Oil	Supplement	OmegaPl...	SUP005	false	4.60	FastShip	Express	1.50	0.01
Bluetooth Speak...	Electronics	BoomBox	SUP008	true	3.80	FastShip	Express	1.50	0.01
Denim Jeans	Fashion	DenimCo	SUP001	false	4.20	SafeLogistics	Standard	3.00	0.02
Running Shoes	Fashion	RunFast	SUP006	false	4.80	SafeLogistics	Standard	3.00	0.02
Smart Watch	Electronics	TimeX	SUP002	true	4.70	EcoDelivery	Eco	5.00	0.05
Phone Case	Electronics	ShieldCa...	SUP008	false	4.40	SafeLogistics	Standard	3.00	0.02
Phone Case	Electronics	ShieldCa...	SUP008	false	4.40	EcoDelivery	Eco	5.00	0.05
Vitamin C 1000...	Supplement	VitaBoost	SUP005	false	3.70	FastShip	Express	1.50	0.01
Microwave	Home_Applian...	HomePro	SUP003	true	4.10	FastShip	Express	1.50	0.01
Denim Jeans	Fashion	DenimCo	SUP001	false	4.20	FastShip	Express	1.50	0.01
Running Shoes	Fashion	RunFast	SUP006	false	4.80	EcoDelivery	Eco	5.00	0.05
Blender	Home_Applian...	BlendIt	SUP003	true	3.90	EcoDelivery	Eco	5.00	0.05
Microwave	Home_Applian...	HomePro	SUP003	true	4.10	FastShip	Express	1.50	0.01
coverage_region text	supplier_name text	contact text	promo_name text	promo_type text	discount_rate numeric (10,2)	start_date timestamp without time zone	end_date timestamp without time zone		
Nationwide	HomeStyle Gro...	02-960-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Bangkok Only	GadgetWorld	02-120-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Nationwide	Wellness Plus	02-206-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Nationwide	GadgetWorld	02-120-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Nationwide	Thai Fashion Co.	02-202-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Nationwide	SportZone Co.	02-171-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Bangkok Only	BangkokTech L...	02-535-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Nationwide	GadgetWorld	02-120-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Bangkok Only	GadgetWorld	02-120-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Nationwide	Wellness Plus	02-206-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Nationwide	HomeStyle Gro...	02-960-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Nationwide	Thai Fashion Co.	02-202-00...	No Promotion	None	0.00	2020-01-01 00:00:00	2030-01-01 00:00:00		
Bangkok Only	SportZone Co.	02-171-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Bangkok Only	HomeStyle Gro...	02-960-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		
Nationwide	HomeStyle Gro...	02-960-00...	Tech Gadget Fest	Campaign	0.15	2025-08-01 00:00:00	2025-08-31 00:00:00		

score_id text	risk_score numeric (10,2)	risk_tier text	scored_at timestamp without time zone	shap_values text	return_id text	return_date timestamp without time zone	return_reason text
SCR02268	0.41	High	2025-08-15 04:00:00	{"rating": 0.1, "history": 0...	RET00601	2025-08-29 04:00:00	Changed Mind
SCR02270	0.69	High	2025-08-15 11:00:00	{"rating": 0.1, "history": 0...	RET00602	2025-08-21 11:00:00	Changed Mind
SCR02279	0.41	High	2025-08-15 23:00:00	{"rating": 0.1, "history": 0...	RET00603	2025-08-21 23:00:00	Better Price Elsewh...
SCR02280	0.34	Medium	2025-08-16 08:00:00	{"rating": 0.1, "history": 0...	RET00604	2025-08-25 08:00:00	Defective
SCR02282	0.44	High	2025-08-16 13:00:00	{"rating": 0.1, "history": 0...	RET00605	2025-08-28 13:00:00	Wrong Item
SCR02283	0.49	High	2025-08-16 14:00:00	{"rating": 0.1, "history": 0...	RET00606	2025-08-30 14:00:00	Defective
SCR02296	0.09	Low	2025-08-17 20:00:00	{"rating": 0.1, "history": 0...	RET00607	2025-08-24 20:00:00	Better Price Elsewh...
SCR02300	0.44	High	2025-08-18 02:00:00	{"rating": 0.1, "history": 0...	RET00608	2025-08-28 02:00:00	Defective
SCR02302	0.29	Medium	2025-08-18 07:00:00	{"rating": 0.1, "history": 0...	RET00609	2025-08-31 07:00:00	Wrong Item
SCR02303	0.19	Low	2025-08-18 08:00:00	{"rating": 0.1, "history": 0...	RET00610	2025-08-22 08:00:00	Defective
SCR02306	0.44	High	2025-08-18 12:00:00	{"rating": 0.1, "history": 0...	RET00611	2025-08-24 12:00:00	Changed Mind
SCR02313	0.44	High	2025-08-19 14:00:00	{"rating": 0.1, "history": 0...	RET00612	2025-08-26 14:00:00	Wrong Item
SCR02316	0.54	High	2025-08-19 18:00:00	{"rating": 0.1, "history": 0...	RET00613	2025-08-25 18:00:00	Better Price Elsewh...
SCR02317	0.54	High	2025-08-19 18:00:00	{"rating": 0.1, "history": 0...	RET00614	2025-08-28 18:00:00	Defective
SCR02321	0.19	Low	2025-08-19 22:00:00	{"rating": 0.1, "history": 0...	RET00615	2025-09-03 22:00:00	Changed Mind

return_scenario text	item_condition text	return_status text	refund_amount numeric (10,2)
Standard Return	Unopened	Completed	2090.00
Standard Return	Unopened	Completed	742.50
Standard Return	Unopened	Completed	712.50
Standard Return	Damaged Packagi...	Completed	792.00
Standard Return	Damaged Packagi...	Completed	1424.00
Standard Return	Unopened	Completed	3383.00
Standard Return	Unopened	Completed	5950.00
Standard Return	Damaged Packagi...	Completed	315.00
Standard Return	Used	Completed	525.00
Standard Return	Defective	Completed	1602.00
Standard Return	Damaged Packagi...	Completed	2090.00
Standard Return	Unopened	Completed	1513.00
Standard Return	Damaged Packagi...	Completed	1492.50
Standard Return	Unopened	Completed	1650.00
Standard Return	Defective	Completed	1760.00

Query ทำ Cross Check Query ดู Production ก่อน Clean ว่าค่านี้ที่ได้จะออกมาเป็นยังไง

เขียน QUERY SQL JOIN TABLE OF 8 TABLES TO 1 TABLE

COMMAND

```
CREATE TABLE order_history_complete_v2 AS
```

SELECT

```
o.order_id,
```

```
o.order_date,
```

```
o.expected_delivery_date,
```

```
o.delivery_date,
```

```
o.customer_id,
```

```
c.customer_name,
```

```
c.customer_phone,
```

```
c.gender,
```

```
c.age,
```

```
c.membership_tier,
```

```
c.preferred_channel,
```

```
c.province,
```

```
c.registration_date,
```

```
c.customer_age_days,
```


o.product_id,
p.product_name,
p.category,
p.brand,
p.is_fragile,
p.product_rating,
p.supplier_id,
s.supplier_name,
s.contact AS supplier_contact,

o.courier_id,
co.courier_name,
co.courier_type,
co.avg_delivery_days,
co.damage_rate,
co.coverage_region,

o.promo_id,
pr.promo_name,
pr.promo_type,

pr.discount_rate AS promo_discount_rate,

pr.start_date AS promo_start_date,

pr.end_date AS promo_end_date,

o.channel_type,

o.payment_method,

o.quantity,

o.unit_price,

o.tier_discount_pct,

o.campaign_discount_pct,

o.total_discount_pct,

o.discount_applied_amount,

o.total_amount,

o.delivery_time_expected_days,

o.delivery_days,

(o.delivery_days - o.delivery_time_expected_days) AS delay_days,

o.is_repurchased_item,

o.order_hour,

o.days_since_last_order,

o.hist_order_count,

o.hist_return_rate,

```
COALESCE(r.return_id, 'None') AS return_id,  
  
r.return_date,  
  
COALESCE(r.return_reason, 'Not Returned') AS return_reason,  
  
COALESCE(r.return_scenario, 'Not Returned') AS return_scenario,  
  
COALESCE(r.item_condition, 'Not Returned') AS item_condition,  
  
COALESCE(r.return_status, 'Not Returned') AS return_status,  
  
COALESCE(r.refund_amount, 0) AS refund_amount,  
  
rs.score_id,  
  
rs.risk_score,  
  
rs.risk_tier,  
  
rs.scored_at,  
  
rs.shap_values,  
  
o.is_returned
```

```
FROM orders o
```

```
LEFT JOIN customers c ON o.customer_id = c.customer_id
```

```
LEFT JOIN products p ON o.product_id = p.product_id
```

```
LEFT JOIN suppliers s ON p.supplier_id = s.supplier_id
```

```
LEFT JOIN couriers co ON o.courier_id = co.courier_id
```

```
LEFT JOIN promotions pr ON o.promo_id = pr.promo_id
```

```
LEFT JOIN returns r ON o.order_id = r.order_id
```

```
LEFT JOIN risk_scores rs ON o.order_id = rs.order_id;
```

เช็คตารางดู (ก่อน JOIN)

```
SELECT 'orders' AS table_name, COUNT(*) AS row_count FROM orders
```

```
UNION ALL
```

```
SELECT 'customers', COUNT(*) FROM customers
```

```
UNION ALL
```

```
SELECT 'products', COUNT(*) FROM products
```

```
UNION ALL
```

```
SELECT 'suppliers', COUNT(*) FROM suppliers
```

```
UNION ALL
```

```
SELECT 'couriers', COUNT(*) FROM couriers
```

```
UNION ALL
```

```
SELECT 'promotions', COUNT(*) FROM promotions
```

```
UNION ALL
```

```
SELECT 'returns', COUNT(*) FROM returns
```

```
UNION ALL
```

```
SELECT 'risk_scores', COUNT(*) FROM risk_scores;
```

	table_name text 	row_count bigint 
1	orders	5000
2	customers	500
3	products	20
4	suppliers	10
5	couriers	3
6	promotions	4
7	returns	1455
8	risk_scores	5000

ตรวจสอบดูว่าข้อมูลครบทั้ง 5000 row หรือไม่ (หลัง JOIN)

```
SELECT 'order_history_complete_v2', COUNT(*) FROM order_history_complete_v2;
```

	?column? text	count bigint
1	order_history_complete_v2	5000

เช็คยอดขายโดยใช้ SUM table orders (ก่อน)

```
SELECT
```

```
    COUNT(*) AS total_orders,
```

```
    SUM(total_amount) AS total_revenue,
```

```
    ROUND(AVG(total_amount), 2) AS avg_order_value,
```

```
    SUM(discount_applied_amount) AS total_discount_given,
```

```
    SUM(is_returned) AS total_returns
```

```
FROM orders;
```

	total_orders bigint	total_revenue numeric	avg_order_value numeric	total_discount_given numeric	total_returns bigint
1	5000	10209126.50	2041.83	1112273.50	1455

เช็คยอดขายโดยใช้ SUM table orders (หลัง)

```
SELECT
```

```
    COUNT(*) AS total_orders,
```

```
    SUM(total_amount) AS total_revenue,
```

```
    ROUND(AVG(total_amount), 2) AS avg_order_value,
```

```
    SUM(discount_applied_amount) AS total_discount_given,
```

```
    SUM(is_returned) AS total_returns
```

```
FROM order_history_complete_v2;
```

	total_orders bigint	total_revenue numeric	avg_order_value numeric	total_discount_given numeric	total_returns bigint
1	5000	10209126.50	2041.83	1112273.50	1455

เช็คทุก Column และ Row

SELECT

COUNT(*) AS total_orders,
 COUNT(customer_name) AS has_customer_name,
 COUNT(membership_tier) AS has_membership_tier,
 COUNT(product_name) AS has_product_name,
 COUNT(category) AS has_category,
 COUNT(supplier_name) AS has_supplier_name,
 COUNT(courier_name) AS has_courier_name,
 COUNT(damage_rate) AS has_damage_rate,
 COUNT(promo_name) AS has_promo_name,
 COUNT(return_id) AS has_return_id,
 COUNT(risk_score) AS has_risk_score

FROM order_history_complete_v2;

	total_orders bigint	has_customer_name bigint	has_membership_tier bigint	has_product_name bigint	has_category bigint	has_supplier_name bigint	has_courier_name bigint	has_damage_rate bigint
1	5000	5000	5000	5000	5000	5000	5000	5000

has_promo_name bigint	has_return_id bigint	has_risk_score bigint
5000	5000	5000

SELECT

channel_type,

COUNT(*) AS total_orders,

SUM(is_returned) AS total_returns,

ROUND(AVG(is_returned) * 100, 2) AS return_rate_pct,

ROUND(SUM(total_amount), 2) AS total_revenue

FROM orders

GROUP BY channel_type

ORDER BY total_revenue DESC;

	channel_type character varying (20) 🔒	total_orders bigint 🔒	total_returns bigint 🔒	return_rate_pct numeric 🔒	total_revenue numeric 🔒
1	Shopee	1250	364	29.12	2643235.00
2	TV_Show	1243	370	29.77	2530556.00
3	Mobile_App	1231	352	28.59	2528231.50
4	TikTok	1276	369	28.92	2507104.00

SELECT

channel_type,

COUNT(*) AS total_orders,

SUM(is_returned) AS total_returns,

ROUND(AVG(is_returned) * 100, 2) AS return_rate_pct,

ROUND(SUM(total_amount), 2) AS total_revenue

FROM order_history_complete_v2

GROUP BY channel_type

ORDER BY total_revenue DESC;

	channel_type character varying (20) 🔒	total_orders bigint 🔒	total_returns bigint 🔒	return_rate_pct numeric 🔒	total_revenue numeric 🔒
1	Shopee	1250	364	29.12	2643235.00
2	TV_Show	1243	370	29.77	2530556.00
3	Mobile_App	1231	352	28.59	2528231.50
4	TikTok	1276	369	28.92	2507104.00

SELECT

province AS "Province",

COUNT(*) AS "Total Orders",

SUM(is_returned) AS "Return Orders",

ROUND((SUM(is_returned)::numeric / COUNT(*)*100, 2) AS "Return Rate (%)"

FROM order_history_complete_v2

GROUP BY province

ORDER BY "Return Rate (%)" DESC;

	Province character varying (20) 🔒	Total Orders bigint 🔒	Return Orders bigint 🔒	Return Rate (%) numeric 🔒
1	Chonburi	641	209	32.61
2	Songkhla	678	215	31.71
3	Chiang Mai	532	156	29.32
4	Nonthaburi	570	167	29.30
5	Khon Kaen	627	180	28.71
6	Remote_Area	803	218	27.15
7	Phuket	696	188	27.01
8	Bangkok	453	122	26.93